# SwiftLog Logistics

## Executive Business Intelligence Report

Prepared by: Agu Precious Ayomide

Track: Data Analytics

## 1. Summary

SwiftLog recorded NGN 14.44 million in net revenue across 5,000 orders from 1,000 customers. Net revenue excludes cancelled orders. Cancellations are the largest single problem in the business: 32.52% of orders were cancelled, and together they account for NGN 7.23 million of the NGN 21.67 million in gross order value.

Delivery performance is the second concern. Only 36.28% of shipments are recorded as on time, and 63.72% are recorded as delayed. Late delivery is also the most common reason for returns, which stand at 9.46% of orders. Delivery reliability therefore affects revenue both directly and through returns.

Revenue itself is in reasonable shape. It is spread across a wide customer base, the top ten customers account for about 3% of net revenue, and 951 of the 1,000 customers have placed more than one order, a retention rate of 95.10%.

## 2. Scope and data

The analysis uses eleven tables supplied for the project: customers, orders, order items, products, suppliers, warehouses, inventory, shipments, delivery performance, employees and returns. The tables were checked for missing values, duplicate keys and broken relationships, and none were found. Every order links to a customer, every order has exactly one shipment, and every return links to a valid order.

Three cleaning decisions shape the figures in this report. Inventory rows that repeated a warehouse and product combination were summed, on the basis that they are separate stock batches. Revenue was calculated as quantity multiplied by the product's current unit price, because the data holds no price history. Suppliers could not be linked to any product or shipment, so supplier performance is not measured.

## 3. Key figures

The figures below are the ones shown on the dashboard. Where a figure could be read two ways, the basis is stated.

| Measure | Value | Basis |
|---|---|---|
| Total orders | 5,000 | All orders in the period |
| Net revenue | NGN 14.44M | Non-cancelled orders |
| Gross revenue | NGN 21.67M | All orders, including cancelled |
| Average order value | NGN 4,279.55 | Net revenue divided by non-cancelled orders |
| Cancellation rate | 32.52% | Cancelled orders as a share of all orders |
| On-time delivery | 36.28% | 1,814 of 5,000 shipments marked On Time |
| Delayed shipments | 63.72% | 3,186 of 5,000 shipments marked Delayed |
| Average delay | 3.57 days | Across all shipments |
| Return rate | 9.46% | 473 orders with at least one return, out of 5,000 |
| Customers | 1,000 | Customers on file |
| Customer retention | 95.10% | 951 customers with more than one order, out of 1,000 |
| Products | 250 | Products in the catalogue |
| Stock held | 1,007,764 units | Across fifteen warehouses |
| Units sold to stock | 0.08 | Units sold divided by units held |

# 4. Findings

1. **Cancellations:** Cancelled orders are the most important operational issue. There are 1,626 shipments linked to cancelled orders and 1,716 linked to pending orders, against 1,658 for delivered orders. A cancelled order should not have a shipment at all, which suggests goods are being dispatched before an order is confirmed. This needs to be checked against the warehouse process before any target is set.

2. **Delivery:** The delayed rate is similar at all three locations: 64.26% in Lagos, 63.60% in Abuja and 62.48% in Kano. On-time performance follows the same pattern, at 35.74% in Lagos, 36.40% in Abuja and 37.52% in Kano. The gap between locations is small, so the problem looks network-wide rather than tied to one site. Lagos carries the most volume, with 2,647 of the 5,000 shipments and 1,701 delayed shipments, so it accounts for the largest share of delays. The delivery data also contains a conflict. 1,203 shipments are marked On Time but show one or more days of delay. The on-time rate therefore depends on which field is used. This report uses the Status field, which records the delivery outcome. Counting only shipments with zero delay days gives an on-time rate of 12.22%, so the true figure may be lower than 36.28%.

3. **Returns:** Late delivery is the most common reason for returns, with 185 of 500 return records. Wrong item accounts for 163 and damaged goods for 152. Return rates also vary by product. P032 has the highest rate at 21.28%, followed by P017 at 18.87%, and both are worth a product review because they are not among the largest sellers.

4. **Inventory:** The business holds about 1.0 million units against 82,442 units sold, a ratio of 0.08. Stock is therefore high relative to demand, which ties up cash. Lagos holds 53.43% of stock units, followed by Abuja at 27.15% and Kano at 19.42%. Fashion has the highest units-sold-to-stock ratio by category, at 0.084, but the three categories are close together.

5. **Customers and products:** Revenue is spread widely. Kano generates the most net revenue, at about NGN 3.9 million, followed by Lagos at 3.7 million, Ibadan at 3.5 million and Abuja at 3.4 million. Average order value per customer is highest in Lagos at 5.2 orders. The largest customer, Customer 223, generated NGN 58,660.46 across eight orders. By product, P085 leads on revenue at NGN 143 thousand and P245 leads on units sold at 487. Fashion is the largest category by net revenue, at NGN 5.4 million.

6. **Demand over time:** Monthly orders ranged from 577 to 654 between January and August. August was the busiest month with 654 orders, followed by April with 638 and May with 612. September shows 154 orders, but the data for that month is partial, so it should not be read as a fall in demand. Nine months is too short to say whether August is a seasonal peak.

# 5. Limitations

- Several limitations affect how far these figures can be relied on.
- Revenue uses current unit prices, so it is an estimate rather than the amount each order actually paid.
- The delivery status conflicts with delay days, as described above.
- Supplier performance cannot be measured without a link between suppliers and products.
- Warehouse capacity is not recorded, so warehouse stock share is a proxy rather than a utilisation rate.
- Average delivery time cannot be calculated because the data has no delivery dates, so average delay is used instead. The data covers nine months, which is not enough to establish seasonality.

## 6. Recommendations

Eleven recommendations follow from these findings and are set out in full in the Part 6 document. The most important are these.

- First, stop dispatching shipments before orders are confirmed, since this affects the largest number of shipments.
- Second, introduce a confirmation check before dispatch to reduce cancellations, with a target of 20% from the current 32.52%.
- Third, tackle late delivery, which is the largest cause of returns.
- Fourth, reduce stock levels on slow-moving lines to release working capital.
- Fifth, link suppliers to products and shipments so that supplier performance can be measured in the next period.

The targets above are suggestions for management to confirm. They are not forecasts drawn from the data.

## 7. Conclusion

SwiftLog's revenue base is healthy, but a third of orders are cancelled and most deliveries are recorded as late. Improving the confirmation and dispatch process and the delivery record would have the largest effect on both revenue and returns. The dashboard supports monitoring these measures monthly, provided the delivery status definition is agreed and applied consistently.